Shorouk Academy.

Department of Computer Science.

Tools and Technologies.

Online Pizza Ordering System.

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Tools and Technologies

The Online Pizza Ordering System will use a set of technologies and software tools to develop the website, manage customer and order information, protect user data, and test the system. These tools are selected to support the main system requirements and make the application easy to maintain and use.

Front-End Technologies

HTML5

HTML5 will be used to create the structure of the website pages, including the home page, pizza menu, registration, login, shopping cart, and order confirmation pages.

CSS3

CSS3 will be used to control the website's appearance, including colors, fonts, layouts, and page elements. It will help create a clear and consistent interface.

JavaScript

JavaScript will be used to add interactive features, such as updating the shopping cart, changing item quantities, calculating prices, and validating information entered by customers.

Bootstrap

Bootstrap will be used to create a responsive design so that the website can be accessed on computers, tablets, and mobile devices.

Backend Technology

Node.js

Node.js is an alternative backend technology that can handle customer requests, communicate with the database, and support the system's ordering operations.

ASP.NET Core

ASP.NET Core is a Microsoft framework that can be used to develop secure web applications and APIs. It can support user authentication, order management, and communication with Microsoft SQL Server.

PHP

PHP will handle the website's main operations, including user registration and login, order processing, price calculations, and communication with the database. It will also support the different permissions assigned to customers, restaurant staff, and administrators.

Backend API

A backend API will allow the website's frontend to communicate with the server to retrieve menu items, submit orders, update shopping carts, and receive order status information.

Database Management System

MySQL

MySQL will store and manage the system's data, including customer accounts, pizza details, prices, menu availability, shopping orders, and order statuses. It will help maintain organized and consistent information.

Microsoft SQL Server

Microsoft SQL Server is an alternative database management system that can store and manage customer information, menu items, order details, and other system data.

UI/UX Design

User Interface (UI) Design

UI design will focus on the visual elements of the website, including colors, fonts, buttons, navigation menus, pizza cards, and shopping cart layouts.

User Experience (UX) Design

UX design will focus on making the website easy to understand and use. Customers should be able to browse the menu, customize pizzas, place orders, and check order status with minimal confusion.

Figma

Figma may be used to create wireframes and interface prototypes before the website is developed. This will help the team plan the page layouts and review the design.

Development Environment

XAMPP

XAMPP will provide a local development environment for running PHP and MySQL on a computer. It will allow the team to develop and test the website before deployment.

Visual Studio Code

Visual Studio Code will be used to write, edit, and organize the website's source code, including HTML, CSS, JavaScript, and PHP files.

Version Control and Team Collaboration

Git

Git will be used to track changes to project files and maintain a history of the development process.

GitHub

GitHub will be used to store the project files online, share updates, and allow team members to collaborate throughout development.

Security Technologies

Password Hashing

Password hashing will be used to store user passwords securely instead of saving them as plain text.

HTTPS/TLS

HTTPS/TLS will protect information transmitted between the user's browser and the website when the application is deployed with HTTPS enabled.

Server-Side Validation

Server-side validation will check information submitted by users before processing it. This will help prevent invalid data from being stored or used in orders.

Role-Based Access Control

Role-based access control will restrict system functions according to each user's role. Customers will manage their own orders, restaurant staff will handle order preparation, and administrators will manage system information and settings.

Testing Tools

Web Browsers

Google Chrome, Mozilla Firefox, and Microsoft Edge will be used to test the website's appearance and functionality across different browsers.

Manual Testing

Manual testing will be used to check registration, login, menu browsing, pizza customization, shopping cart operations, price calculations, order placement, and order status updates.

Database Testing

Database testing will check whether customer information, menu items, and order details are stored, updated, and retrieved correctly.

Documentation Tools

LibreOffice Writer

LibreOffice Writer will be used to prepare, edit, and format the project report and other documentation.

PDF

PDF format will be used to submit and share the finalized report while preserving its formatting.